Clinical Study Summary Report (CSSR)
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1. Administrative & Study Identification

Full Study Title: An Open-label, Phase 1 Study of the Safety, Tolerability, and Pharmacokinetics of KarXT and Dual-burst Release of Xanomeline With Immediate-release Trospium Chloride in Adolescents With Psychiatric Disorders
Short Title/Acronym: KarXT Adolescent PK/Safety Study
Protocol Number: BMS-986519 / BMS-986510
NCT Number: NCT06853171
Sponsor Name: Bristol-Myers Squibb
Investigational Product: KarXT (Xanomeline and Trospium Chloride IR Capsule) and KarX-EC (Xanomeline Enteric-coated)
Phase of Development: Phase 1
Medical Monitor: Bristol-Myers Squibb Clinical Operations Lead

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2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the safety and tolerability of multiple doses and ratios of xanomeline and trospium chloride in an immediate-release capsule (KarXT) and dual-burst release of xanomeline with immediate-release trospium chloride in adolescents with psychiatric disorders. **Secondary Objectives:** To evaluate the pharmacokinetics (PK) of multiple doses and ratios of KarXT and KarX-EC, including maximum plasma concentration (Cmax) and area under the concentration-time curve (AUC), in adolescents. **Optimization Target:** Safety, Tolerability, and PK parameter mapping in adolescents.

2.2 Background & Rationale To evaluate the safety, tolerability, and pharmacokinetic profile of the novel muscarinic agonist combination KarXT (Cobenfy) and KarX-EC in an adolescent population diagnosed with a **Primary Condition:** Psychiatric Disorders (Schizophrenia, Bipolar, ADHD, Tourette's, ASD). The study is necessary to inform dosing regimens and confirm the safety profile in younger patients prior to larger efficacy trials.

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3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Management Pathway:** Phase 1 Open-label PK & Safety characterization **Control Type:** Single-arm/Open-label cohorts **Blinding:** Open-label **Randomization:** N/A (Assigned to respective dose cohorts)

3.2 Planned Enrollment & Duration Target N: 24 evaluable participants **Number of Sites:** Multiple sites in the United States (e.g., Arkansas, California, Georgia, Kansas) **Estimated Study Start:** 29-Apr-2025 (Actual) **Estimated Primary Completion:** 22-Jul-2025 (Actual)

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4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. LAR (legal guardian or caregiver) must have signed and dated an IRB/IEC-approved ICF in accordance with regulatory, local, and institutional guidelines. 2. **Diagnosis Threshold:** Confirmed Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition, Text Revision (DSM-5-TR) psychiatric diagnosis of 1 of the following: Schizophrenia or schizoaffective disorder, Bipolar I or II disorder, Attention-deficit/ hyperactivity disorder (ADHD), Tourette's disorder, or Autism spectrum disorder (ASD). 3. Participant is judged by the investigator to be clinically stable (e.g., no psychiatric hospitalization within the last 6 months; no imminent risk of suicide or injury to self, others, or property).

4.2 Exclusion Criteria 1. Any clinically significant neurological, metabolic (including type 1 diabetes), hepatic, renal, hematological, pulmonary, cardiovascular, GI (including active obstructive GI disorders), carcinoma, active biliary disorders (eg, symptomatic gallstones) and/or urological disorder, congestive heart failure (uncontrolled), or CNS infection. 2. Participant has a risk for suicidal behavior during the study, as determined by the investigator's clinical judgment and C-SSRS. 3. eGFR < 60 mL/min; History of Gilbert's Disease or history of liver disease (Child-Pugh class A and higher); History or high risk of urinary retention, gastric retention, or narrow-angle glaucoma; Participants with history of bladder stones or recurrent UTIs.

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5. Intervention & Treatment Plan

5.1 Investigational Regimen Medical Methodology: Muscarinic Agonist Combination Therapy **Dosage/Frequency:** Multiple doses and ratios of xanomeline and trospium chloride evaluated across distinct cohorts on specified days. **Route of Administration:** Oral (Capsule and Enteric-coated). **Duration of Treatment:** Acute dosing up to Day 15 for PK evaluation, with safety follow-up extending to Day 43 (Total Duration: Up to 43 Days).

5.2 Concomitant & Prohibited Therapy Permitted: Allowed maintenance medications for stable psychiatric conditions not interfering with cholinergic/muscarinic pathways. **Prohibited:** Medications carrying high anticholinergic burden or known to significantly alter gastrointestinal transit/urinary retention.

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6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures | | :--- | :--- | :--- | | **Screening** | Day -28 to 0 | Informed Consent (LAR), Assent, DSM-5-TR Psychiatric Diagnosis Confirmation, Medical History, C-SSRS | | **Baseline** | Day 0 | Cohort Assignment, First Dose, Baseline Vital Signs, Baseline Labs | | **Interim** | Days 1 to 15 | Intensive PK Blood Draws (Cmax, Tmax, AUC), Safety Assessments. **Onsite Visit Cadence:** High frequency (Days 1-15) for intensive PK sampling. **Checklist Review Interval:** Daily dosing and AE diary reviews through Day 15. | | **End of Study** | Up to Day 43 | Final Vital Signs, Exit Interview, IP Accountability, Final AE/SAE monitoring. **Remote Monitoring Frequency:** Weekly follow-ups from Day 15 to Day 43. |

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7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Number of participants experiencing Adverse Events (AEs), Serious Adverse Events (SAEs), and AEs of Special Interest (AESIs). **Measurement Method:** Clinical investigator observation, patient reporting, and standardized safety tracking forms from Day 1 up to Day 43.

7.2 Secondary & Exploratory Endpoints * Maximum observed plasma concentration (Cmax) and Time of maximum observed plasma concentration (Tmax) up to Day 15. * Area under the concentration-time curve in 1 dosing interval (AUC(TAU)) and AUC from time zero to time of last quantifiable concentration (AUC(0-T)) up to Day 15. * Concentration at the end of a dosing interval (Ctau), Cmax accumulation index (AI_Cmax), and AUC accumulation index (AI_AUC) up to Day 15.

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8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring collected via clinician assessment and patient/LAR reports up to Day 43. **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness, tracked meticulously up to Day 43. **Data Monitoring Committee (DMC):** Internal safety evaluation committee established by Bristol-Myers Squibb to review cohort PK and safety data.

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9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all participants receiving ≥ 1 dose) and Pharmacokinetic (PK) Set. **Sample Size Justification:** Target $N=24$; adequately powered for an exploratory Phase 1 PK, safety, and tolerability assessment. **Handling of Missing Data:** Missing PK data excluded from non-compartmental analysis; descriptive statistics utilized for safety parameters.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** High-frequency onsite monitoring due to intensive PK sampling requirements. **Education Protocol (HCP):** Rigorous site training on intensive PK sampling timing and processing. **Education Protocol (Patient):** LAR consent and adolescent assent procedures, dosing compliance instructions. **Quality Audit Mechanism:** Pharmacokinetic assay validation and adherence to AE/SAE reporting timelines. **Audit Trail:** eCRF locking and comprehensive data clarification form workflows managed by the sponsor.

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11. Study Identifiers & Governance

Primary ID: {{BMS-986519}} **Registry Number:** {{NCT06853171}} **Sponsor/Lead Organization:** {{Bristol-Myers Squibb}} **Study Title:** {{KarXT Adolescent PK/Safety Study}} **Official Regulatory Title:** {{An Open-label, Phase 1 Study of the Safety, Tolerability, and Pharmacokinetics of KarXT and Dual-burst Release of Xanomeline With Immediate-release Trospium Chloride in Adolescents With Psychiatric Disorders}}

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12. Clinical Framework (Psychiatric Specific)

Primary Condition: {{Psychiatric Disorders (Schizophrenia, Bipolar, ADHD, Tourette's, ASD)}} **Diagnosis Threshold:** {{Confirmed DSM-5-TR psychiatric diagnosis}} **Medical Methodology:** {{Muscarinic Agonist Combination Therapy}} **Optimization Target:** {{Safety, Tolerability, and PK parameter mapping in adolescents}}

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13. Study Procedures & Methodology

Management Pathway: {{Phase 1 Open-label PK & Safety characterization}} **Education Protocol (HCP):** {{Rigorous site training on intensive PK sampling timing and processing}} **Education Protocol (Patient):** {{LAR consent and adolescent assent procedures, dosing compliance instructions}} **Quality Audit Mechanism:** {{Pharmacokinetic assay validation and adherence to AE/SAE reporting timelines}}

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14. Observation & Follow-Up Schedule

Total Duration: {{Up to 43 Days}} **Onsite Visit Cadence:** {{High frequency (Days 1-15) for intensive PK sampling}} **Remote Monitoring Frequency:** {{Weekly follow-ups from Day 15 to Day 43}}

Checklist Review Interval: {{Daily dosing and AE diary reviews through Day 15}}

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15. Sign-off & Approval

Role	Name	Signature	Date
Principal Investigator	[Name]	____	[DD-MMM-YYYY]
Clinical Operations Lead	[Name]	___	[DD-MMM-YYYY]
Lead Statistician	[Name]	_____	[DD-MMM-YYYY]